

Lab Experiment 1

Subject - Machine Learning Theory and Practice

Subject Code - E1CAB 613

1. Dataset

Cafe Sales – Dirty Data for Cleaning Training

<https://www.kaggle.com/datasets/ahmedmohamed2003/cafe-sales-dirty-data-for-cleaning-training>

2. Objective

To apply data cleaning and transformation techniques on a real-world messy dataset using Python.

3. Lab Question

You are given a CSV file containing cafe sales transaction data. Perform the following tasks using Python.

Tasks & Marks Distribution (Total: 10 Marks)

Task 1: Create a Virtual Environment and Install Required Packages (1 Mark)

- Create a Python virtual environment for this lab experiment.
- Activate the virtual environment.
- Install the required libraries:
 - pandas
 - numpy
 - Matplotlib (or seaborn)
- Show the command(s) used.

Purpose: To ensure a clean and isolated Python environment.

Task 2: Create a Load and Inspect the Dataset (1 Mark)

- Load the CSV file into a pandas DataFrame.
- Display:
 - First 5 rows

- Last 5 rows
- Data types and summary information of the dataset.

Purpose: To understand the structure and quality of the data.

Task 3: Handle Missing Values (2 Marks)

- Identify columns containing missing values.
- Handle missing values using appropriate techniques:
 - Numerical columns → mean or median
 - Categorical columns → suitable placeholder (e.g., "Unknown")
 - Date columns → valid date handling method

Purpose: To ensure completeness of the dataset.

Task 4: Remove Duplicate Records (2 Marks)

- Identify duplicate rows in the dataset.
- Remove duplicate records.
- Report the number of duplicate rows removed.

Purpose: To avoid data redundancy and bias.

Task 5: Detect Outliers (2 Marks)

- Detect outliers in a numerical column using:
 - Interquartile Range (IQR) method **OR**
 - Box plot visualisation.
- Display the outlier records.

Purpose: To identify abnormal or extreme values.

Marks Distribution Summary

Task	Description	Marks
1	Create a Virtual Environment and Install Required Packages	1
2	Data loading & inspection	1
3	Handling missing values	2
4	Removing duplicates	2
5	Outlier detection	2
	Timely Submission	2
Total		10

General Instructions

1. Ensure that your code includes relevant comments to enhance readability and understanding. Subsequently, upload your code to GitHub for version control and collaborative access.
2. Include descriptive comments within the code, explaining its functionality and logic.
3. Attach a PDF document named "your_register_number_Lab_No.pdf" to the submission. The PDF document should include screenshots of the code and the output screen.
4. Kindly attach the Github link to your lab exercise as well.